0.1 Introduction

Consider 2D Pure heat diffusion on a 2D cartesian domain . We want to find the temperature distribution on this domain using finite volume method.
The Governing equation for this transport is:

$$\frac{\partial}{\partial x}(k.\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k.\frac{\partial T}{\partial y}) = 0 \tag{1}$$

$$\nabla \cdot (k\nabla T) = 0 \tag{2}$$

1. First we discretise the domain into a meshgrid with NxN grid points, around each grid we define a finite volume which boundaries(faces) should lie midway between the neighbouring gridpoints.

2. Integrate the governing equation on those finite volume.

$$\int \nabla \cdot (k\nabla T)\, dV = 0 \tag{3}$$

Using gauss Theorem we convert this volume integral of a divergence into a surface integral.

$$\int \hat{n} \cdot (k\nabla T)\, dA = 0 \tag{4}$$

Note: for this finite volume only 4 faces are involved in the transport since its a 2D transport : e face , w face , s face ,n face, $\hat{n}$ is the normal vector to those faces and is directed outwards.

$$\int \hat{n}_e \cdot (k\nabla T)\, dA_e + \int \hat{n}_w \cdot (k\nabla T)\, dA_w + \int \hat{n}_n \cdot (k\nabla T)\, dA_n + \int \hat{n}_s \cdot (k\nabla T)\, dA_s = 0 \tag{5}$$

$$kA\frac{\partial T}{\partial x}\bigg|_e - kA\frac{\partial T}{\partial x}\bigg|_w + kA\frac{\partial T}{\partial y}\bigg|_n - kA\frac{\partial T}{\partial y}\bigg|_s = 0 \tag{6}$$

3. This term $kA\frac{\partial T}{\partial x}\big|_e$ is simply the diffusive flux of Heat across east face : How much heat is being transported by diffusion across the east face .

4. Now what we are left is to decide how T (our function) varies between the grid points, Lets assume T varies piecewise Linear ,then :

   (a)
   $$\frac{\partial T}{\partial x}\bigg|_e = \frac{T_E - T_P}{h_{PE}} \tag{7}$$

   (b)
   $$\frac{\partial T}{\partial x}\bigg|_w = \frac{T_P - T_W}{h_{WP}} \tag{8}$$

   (c)
   $$\frac{\partial T}{\partial y}\bigg|_n = \frac{T_N - T_P}{h_{PN}} \tag{9}$$

(d)

$$\left.\frac{\partial T}{\partial y}\right|_s = \frac{T_P - T_S}{h_{SP}} \tag{10}$$

5. Substitute Back into EQN 6 :

$$k_e A_e(\frac{T_E - T_P}{h_{PE}}) - k_w A_w(\frac{T_P - T_W}{h_{WP}}) + k_n A_n(\frac{T_N - T_P}{h_{PN}}) - k_s A_s(\frac{T_P - T_S}{h_{SP}}) = 0 \tag{11}$$

6. Rearranging gives :

$$(\frac{k_e A_e}{h_{PE}} + \frac{k_w A_w}{h_{WP}} + \frac{k_n A_n}{h_{PN}} + \frac{k_s A_s}{h_{SP}})T_P = \frac{k_e A_e}{h_{PE}}T_E + \frac{k_w A_w}{h_{WP}}T_W + \frac{k_n A_n}{h_{PN}}T_N + \frac{k_s A_s}{h_{SP}}T_S = 0 \tag{12}$$

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + a_S T_S \tag{13}$$

7. Once the iterative formula EQN (13) has been obtained we have to solve it on each grid point to determine the temperature distribution , this process is done numerically with python through an iterative process and a convergence criterion.

   The code is presented in the files

0.2 Parameters of the simulation

1. The domain is a square of length $2m$
2. N is the number of grid points for N xN lattice , N=50 , so the total number of grid points for the simulation is 2500
3. thermal conductivity $k = 0.1$
4. Cross-sectional area of each faces e,w,n,s is A =0.001
5. INITIAL BOUNDARY CONDITION , T= 1 on North and West side of domain.
6. Grid spacing is the same everywhere , so $h_{PE}, h_{WP}, h_{PN}, h_{SP} = h$,

$$h = \frac{L}{N-1} = \frac{1}{12} \tag{14}$$

0.3 RESULTS OF SIMULATION

The convergence criterion is set to 10^{-8}, when the numercial error i.e the sum of differences between the actual and previous value of Temperature at the grid points become $> 10^{-8}$ it implies that the solution has converged. A plot of the numerical error is done against the iteration number for every 1000 iterations.Some of the plots are shown below the rest is in the files provided with this report.

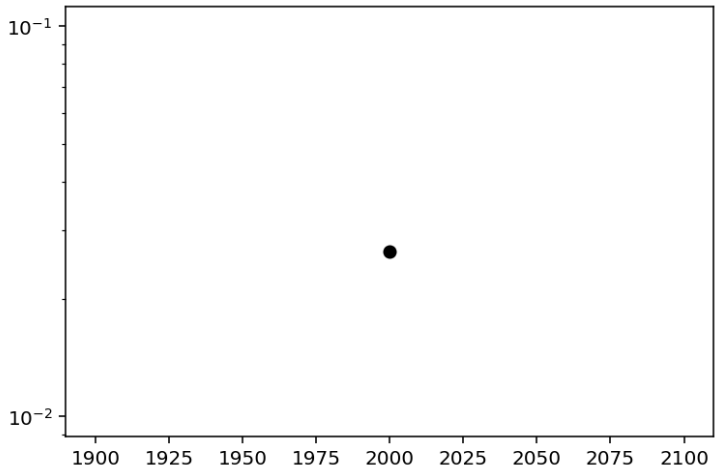


Figure 1:

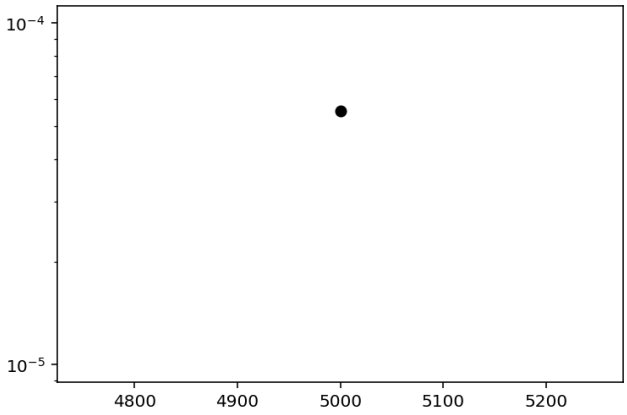


Figure 2:

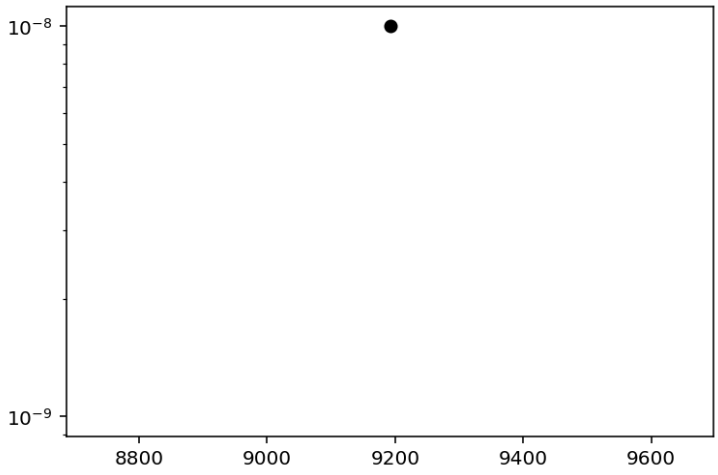


Figure 3:

The 2D contour plot of the temperature distribution is displayed below:

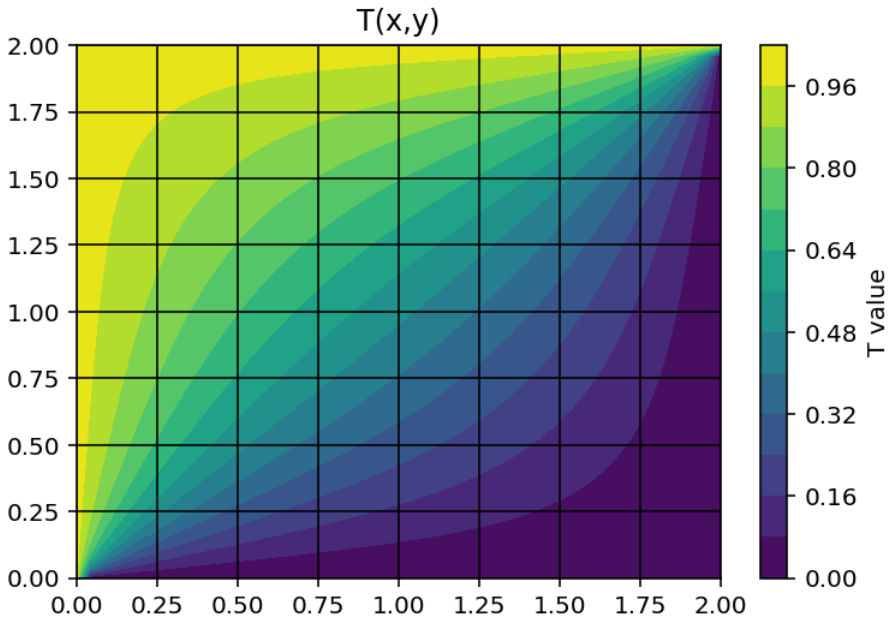


Figure 4:

0.4 Discussion of Results

For a 2D pure diffusion problem (no convection term)
The isotherms are smooth and symmetric because the boudary conditions and domain are symmetric , also since we have pure diffusion there is no biased direction for heat flow.